

ANUP PAUL ELIAS

PROCESS
ENGINEERING
LEADER

22+ years | Refining | Petrochemicals | EPC / FEED | Commissioning | Optimization | Process Safety

Chennai, India | a.p.elias@gmail.com | anuppaul007.github.io

EXECUTIVE PROFILE

Chemical / Process Engineering leader combining owner-operator judgement with EPC delivery experience across the full asset lifecycle. Experience spans conceptual development, FEED, process design, plant support, commissioning and start-up, debottlenecking, troubleshooting, process safety, technical assurance and engineering-team leadership. Technical exposure includes hydrotreating, catalytic reforming, PDH, ethylene oxide, propylene oxide / glycol, acetic acid, VAM, n-paraffins, utilities and offsites.

CAREER IMPACT

22+

Years across process engineering, operations and project delivery

15

Process engineers led in a major engineering organization

+19%

Annual Acetic Acid production increase from debottlenecking

18->92%

Wastewater plant run-time efficiency improvement

1.36->1.9

MMTPA NHT capacity increase through revamp leadership

3

International technical conference contributions

PROFESSIONAL EXPERIENCE

Tecnicas Reunidas, Chennai - Process Engineering Leader

2026 - Present

- Lead process engineering delivery in an EPC environment with focus on technical quality, execution discipline, interfaces and team capability.
- Mentor process engineers and support consistent engineering practices, deliverable quality and technical decision-making.

Farabi Yanbu Petrochemical Company, Saudi Arabia - Senior Process Engineer / Process Engineering Lead

2023 - 2026

- Monitored and improved performance of hydrotreating, UOP Molex and Sour Water Stripping systems; led troubleshooting, benchmarking and optimization support.
- Provided process-safety and reliability support and led a small process engineering team supporting n-paraffin / LAB operations.

Dow Chemical Company, India / USA support - Process Engineering Manager

2020 - 2023

- Led 15 process engineers delivering CAPEX, OPEX, plant-support and optimization work for major US manufacturing sites.
- Managed process engineering development of an Ethylene Oxide FEED package and served as engineering coach / subject-matter resource.
- Supported overpressure-compliance and site-project execution, including PDH / olefins-related assignments.

Sadara Chemical Company, Saudi Arabia - Senior Process Improvement Engineer

2014 - 2019

- Led process-improvement support across chemical units and port facilities, covering operability, reliability, safety, technical feasibility and vendor review.
- Supported MOC, HAZOP, process-hazard assessment, commissioning / start-up and major operating-asset improvement initiatives.

EARLIER EXPERIENCE

Saudi International Petrochemical Company (SIPCHEM), Saudi Arabia - Process Engineer

2010 - 2014

- Led plant troubleshooting, optimization, RCA, bottleneck identification, guarantee-test support and capital-project coordination.
- Directed wastewater reliability improvement that increased run-time efficiency from 18% to 92%.
- Implemented Acetic Acid debottlenecking changes that increased annual production by 19%.

Dow Chemical Company, India - Process Engineer

2007 - 2010

- Process engineering support for global chemical operations and projects.

Essar Oil Ltd., India - Process Engineer

2005 - 2007

- Lead process support for NHT revamp, increasing capacity from 1.36 to 1.9 MMTPA, followed by commissioning, start-up and stabilization of NHT / CCR operations.

Tuticorin Alkali and Fertilizers Ltd., India - Operation Engineer

2004 - 2005

- Early-career operating-plant experience in continuous chemical production.

TECHNICAL EXPERTISE

Process: FEED, process design, plant support, debottlenecking, troubleshooting, commissioning, technical assurance, process safety, HAZOP / LOPA / PHA, MOC, relief / flare concepts.

Tools: Aspen HYSYS, Aspen Plus, PRO/II, HTRI, Aspen EDR, AFT Fathom, FloMaster, PHAST, HAZOP Manager.

Technology: NHT (Axens), KHT (UOP Unionfining), UOP Molex, CCR / Aromizing, UOP Oleflex PDH, EO, PO, PG, Acetic Acid, VAM.

EDUCATION

Postgraduate in General Management (PGGM)
Indian Institute of Management Indore, 2024

MBA
Alagappa University, 2006

B.Tech - Chemical Engineering
University of Madras, 2004

SELECTED TRAINING & CERTIFICATIONS

Certified Auditor - ISO 9001:2015, ISO 14001:2015, ISO 45001:2018 & ISO 50001:2018 - DNV, 2024

ISO 50001:2018 Energy Management System - Implementer - DNV, 2024

Loss Prevention Principles - Sadara Process Safety Risk Management, 2018

Layers of Protection Analysis (LOPA) - Sadara Process Safety Risk Management, 2018

PHAST Dispersion Modelling - DNV-GL, 2015

Petrochemical Heat Exchanger Design - HTRI, 2014

TECHNICAL PRESENTATIONS

Plant Layout for Inherently Safe Design of a Petrochemical Facility - Gulf Downstream Association International Conference, 2018

Designing and Modeling Dividing Wall Columns - 4th Middle East Process Engineering Conference, 2017

Thermal Expansion on a Non-blocked Pipeline - Case Study - CCPS Global Process Safety Conference, 2016

PROFESSIONAL AFFILIATIONS

AIChE - Member | IChemE - Associate Member | IChE - Associate Member

Public portfolio: <https://anuppaul007.github.io/> | **Email:** a.p.elias@gmail.com

This public CV intentionally omits date of birth, phone numbers and home address. Detailed credential evidence is available through the online portfolio.